

Abhinav Singh

Targeting Summer 2027 internships in thermal systems, hardware design, and test and validation

765-543-5317 | abhinavsingh1176@gmail.com | linkedin.com/in/abhinav-bhanupratap-singh | Portfolio

EDUCATION

Purdue University

Bachelor of Science, Mechanical Engineering | GPA: 3.30/4.00

West Lafayette, IN

Aug. 2024 – May 2028

- Dean's List. Coursework: Thermodynamics, Fluid Mechanics, Heat Transfer, Systems Measurement & Control, Mechanics of Materials, Statics, Dynamics, Materials Science, Manufacturing Processes.

EXPERIENCE

Combustion Performance & Emissions Engineer Co-op

June 2026 – Dec. 2026

Cummins Inc., Power Systems, Thermal & Fluid Sciences

Seymour, IN

- Assigned engineer on an EPA Tier 4 emissions certification audit for a high-horsepower industrial diesel platform, owning install review, test cell readiness, instrumentation planning, verification, and compliance documentation across Lab Operations, Aftertreatment, and supplier teams.
- Own the genset test phase of a cross-site turbocharger compressor surge investigation, authoring test plans, coordinating weekend test cell runs with technicians, and analyzing compressor behavior against measured data to isolate the operating conditions driving surge.
- Audited data logger channel health across multi-channel engine shutdown datasets in MATLAB, delivering a channel matrix workbook and run condition summary that validated data integrity before downstream cross-site analysis.
- Designed and built a self-initiated desktop search tool indexing a multi-terabyte engineering documentation repository into a SQLite catalog with a natural language front end; approved by engineering leadership and now advancing to a team pilot.

Heat Transfer Undergraduate Researcher

Apr. 2025 – Jan. 2026

Cooling Technologies Research Center (CTRC), Purdue University

West Lafayette, IN

- Built end to end thermal simulation pipelines in ANSYS Fluent and Mechanical, covering geometry, meshing, solver setup, and post-processing, to characterize micro pin-fin heatsink performance under high heat flux for electronics cooling applications.
- Cut simulation turnaround roughly 35% by validating reduced-order 2D and symmetry models against full 3D baselines, preserving solution accuracy while lowering computational cost.

Mechanical Engineering Research Intern

Jul. 2023 – Aug. 2023

Dept. of Mechanical Engineering, Maharaja Sayajirao University

Gujarat, India

- Ran thermal conductivity and sensor sensitivity experiments for solar concentrator and thermocouple heat flux sensor research; led data acquisition and contributed to a thesis manuscript and IP filing alongside faculty and graduate researchers.

LEADERSHIP & PROJECTS

Battery & Thermals Subteam | Purdue Electric Racing

2026 – Present

- Model passive heat rejection paths out of the tractive system accumulator, accounting for conduction through structural mounts, external convection, and radiation to size pack-level thermal margin.

Powertrains Member | ASME Racing, Purdue University

Jan. 2025 – May 2026

- Designed Purdue's first in-house CVT mount in Autodesk Fusion, coordinating with chassis and engine teams to optimize packaging, weight, and cost.
- Designed and fabricated sheet metal manufacturing fixtures, drivetrain shrouds, and adjustable exhaust mounts via manual machining; mitigated heat-related failure modes through root cause analysis.

Board of Directors, Professional Development | Purdue Alumni Student Experience

Sept. 2025 – Present

- Planned and ran professional development events including a mock career fair, networking dinner, and resume reviews, connecting 150+ students with alumni and industry professionals.

CAD, GD&T, and Manufacturing Projects | ME 264, ME 29700GC Coursework

Jan. 2025 – Dec. 2025

- Designed a 17-inch steel wheel rim to OEM specification and a four-cylinder exhaust manifold in Siemens NX, optimizing runner geometry for airflow and thermal performance and producing full GD&T manufacturing drawings.
- Designed a multi-component assembly in Siemens NX applying DFM and DFA principles; machined prototype parts on manual mill and lathe, holding specified tolerances on manufacturing-ready drawings.

Electronics Team | EPICS, Global Air Quality Trekkers

Aug. 2024 – May 2025

- Built the Arduino control system for a portable air purification device, cutting breadboard footprint 50%, refining PID control logic, and validating performance experimentally with the mechanical team.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Siemens NX, Autodesk Fusion, AutoCAD, GD&T, DFM/DFA, tooling and fixture design, engineering drawings

Simulation & Analysis: ANSYS Fluent (CFD), ANSYS Mechanical (FEA), MATLAB, Simulink, GT-Power

Test & Data: Test plan development, DAQ and instrumentation, thermocouples and accelerometers, AVL Concerto, Calterm, root cause analysis, Python data analysis

Fabrication: CNC, manual mill and lathe, waterjet, sheet metal, welding, 3D printing, bandsaw

Programming & Tools: Python, SQL, C++, Excel, GitHub, LaTeX